

MCGREGOR, TX, USA

WMO: 722563

Lat: 31.485N

Lon: 97.316W

Elev: 592

StdP: 14.38

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53952

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.7	27.7	10.0	9.4	30.4	14.8	11.9	34.8	26.7	59.6	24.4	56.1	8.5	0	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.8	101.7	74.6	99.6	74.8	97.7	74.9	78.5	92.1	77.9	91.7	77.2	90.9	10.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	135.5	82.6	74.6	132.2	82.3	73.4	126.8	81.8	42.6	92.4	41.7	91.9	40.9	91.0	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.5	20.5	18.6	DB	17.8	104.8	4.3	2.9	14.7	106.9	12.1	108.6	9.7	110.2	6.6	112.4	(4)
			WB	16.0	80.1	3.7	1.3	13.3	81.0	11.1	81.8	9.0	82.5	6.3	83.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	67.3	48.0	51.9	58.4	66.7	74.6	81.8	85.4	78.7	68.3	58.0	49.3		
		DBStd	15.42	10.07	10.41	9.44	7.40	6.31	4.24	3.48	3.85	6.12	7.95	9.29	9.60	(7)
		HDD50	451	159	93	34	1	0	0	0	0	1	32	131		
		HDD65	2032	531	380	243	68	9	0	0	3	59	245	494		
		CDD50	6754	98	146	295	502	764	953	1088	1097	861	569	271	110	(10)
		CDD65	2862	5	13	39	119	308	503	623	632	414	163	35	8	(11)
		CDH74	33094	29	101	257	979	2934	5884	8111	8378	4597	1549	234	41	(12)
		CDH80	17099	2	19	38	264	1206	3066	4679	4888	2328	564	42	3	(13)
(14)	Wind	WSAvg	9.0	9.2	9.8	10.4	10.5	9.6	8.9	8.3	7.9	8.4	8.7	8.7		
(15)	Precipitation	PrecAvg	36.7	2.6	2.8	3.7	2.9	4.4	3.6	2.0	2.6	3.1	4.1	3.1	3.1	(15)
(16)		PrecMax	48.3	5.4	6.8	9.8	6.7	12.8	10.6	5.6	8.8	9.4	11.0	10.2	8.7	(16)
(17)		PrecMin	25.2	0.6	0.2	1.4	0.6	0.7	0.7	0.0	0.5	0.6	0.5	0.7	0.7	(17)
(18)		PrecStd	6.7	1.6	1.9	1.9	1.7	2.6	2.4	1.5	2.8	2.3	2.7	2.5	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	78.0	82.1	83.9	90.8	96.9	101.8	104.1	105.9	102.1	92.7	84.3	79.1		
(20)		0.4% MCWB	61.4	63.2	66.3	70.1	73.3	74.4	74.5	74.9	73.2	72.4	69.6	64.7		
(21)		2% DB	72.7	76.9	80.6	86.2	92.7	98.9	100.9	101.8	97.4	90.0	80.7	73.3		
(22)		2% MCWB	60.9	62.0	65.7	68.8	73.7	74.9	74.5	75.1	74.0	71.0	66.8	63.1		
(23)		5% DB	69.5	72.5	76.7	82.3	90.1	96.5	99.3	99.8	94.5	86.3	75.5	70.1		
(24)		5% MCWB	60.0	61.4	64.1	68.1	73.4	75.1	74.8	75.0	73.8	69.6	64.5	61.5		
(25)		10% DB	64.5	69.6	72.9	80.3	87.6	92.9	97.2	98.5	91.2	82.3	72.7	65.9		
(26)		10% MCWB	57.3	61.2	62.6	66.8	72.6	74.9	75.0	75.1	72.8	67.3	63.9	59.1		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	67.0	68.9	70.7	75.2	78.5	79.8	79.2	80.1	78.6	76.8	72.8	68.8		
(28)		0.4% MCDB	72.3	74.8	79.0	84.4	87.7	91.8	93.4	93.6	91.5	86.5	80.5	74.2		
(29)		2% WB	64.6	66.6	68.5	72.8	76.8	78.2	78.2	78.4	77.4	74.9	70.0	66.2		
(30)		2% MCDB	69.3	72.2	75.8	81.9	87.0	91.2	92.7	93.0	89.5	83.9	76.3	71.6		
(31)		5% WB	62.0	64.4	66.9	71.0	75.4	77.3	77.5	77.7	76.3	73.1	67.6	63.3		
(32)		5% MCDB	66.5	69.9	73.5	79.7	85.5	90.5	92.1	92.5	87.7	81.1	73.7	67.9		
(33)		10% WB	58.5	61.7	65.1	69.1	74.0	76.4	76.8	76.9	75.2	71.1	65.6	59.6		
(34)		10% MCDB	64.6	68.4	71.2	76.8	84.0	88.8	91.1	91.4	86.1	79.0	71.8	65.0		
(35)	Mean Daily Temperature Range	MDBR	21.2	21.2	21.3	22.3	20.7	21.2	22.6	22.8	22.5	22.9	20.8	20.7		
(36)		5% DB MCDBR	26.3	25.8	23.6	24.2	23.8	24.9	25.6	25.9	25.5	25.0	23.7	25.3		
(37)		5% WB MCWBR	17.1	16.3	13.8	12.8	9.6	7.5	7.0	7.5	9.0	12.0	14.1	17.2		
(38)		5% DB MCDBR	20.1	21.0	19.4	18.9	19.7	20.5	21.9	22.3	21.0	19.4	18.8	20.5		
(39)		5% WB MCWBR	15.9	15.7	13.2	11.4	9.5	7.9	7.0	7.0	8.0	10.6	13.3	16.4		
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.335	0.358	0.399	0.429	0.431	0.443	0.436	0.401	0.370	0.347	0.323		
(41)		taud	2.509	2.463	2.413	2.322	2.296	2.325	2.326	2.332	2.375	2.447	2.456	2.493		
(42)		Ebn at Noon	290	293	293	283	274	272	268	269	275	277	275	280		
(43)		Edh at Noon	28	32	36	41	42	41	41	40	37	32	29	27		
(44)	All-Sky Solar Radiation	RadAvg	940	1089	1425	1722	1833	2086	2133	1974	1654	1321	990	826		
(45)		RadStd	98	185	123	154	163	168	153	119	138	196	121	113		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page